MQTT connection with HiveMQ Broker

Application Note

80617NT12134A Rev.0
2024-03-12
Released
Public

Contents

1 Applicability Table

Table 1: Applicability Table

Products
ME310G1 SERIES
ME910G1 SERIES
ML865G1 SERIES

2 Introduction

2.1 Scope

The scope of this document is to give information about MQTT functionality supported by the Telit modules. Specifically, it is a guide on how to establish an MQTT connection via AT command to the HiveMq free server.

2.2 Audience

This document is intended for system integrators that are using the Telit XX123Z4 module in their products.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: https://www.telit.com/contact-us/

Product information and technical documents are accessible 24/7 on our website:
https://www.telit.com

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk: Notifies the user to take proper grounding precautions before handling the product.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and conditions

Refer to https://www.telit.com/hardware-terms-conditions/.

2.6 Disclaimer

3 MQTT Quick start

3.1 Initial configuration

The module provides three different ports that allow to communicate via AT commands: USB, main UART, and aux UART.

Once you connect one of these three ports on your PC, you can use a generic serial communication terminal software to communicate with the module.

In this guide, the Telit Cinterion AT Controller will be used.

Telit Cinterion AT controller

Telit AT controller allows to establish a serial communication and to send AT commands to the Telit Cinterion module.

As shown in Figure 1, it is possible to write AT commands manually or choose from the default sets present in the left panel.

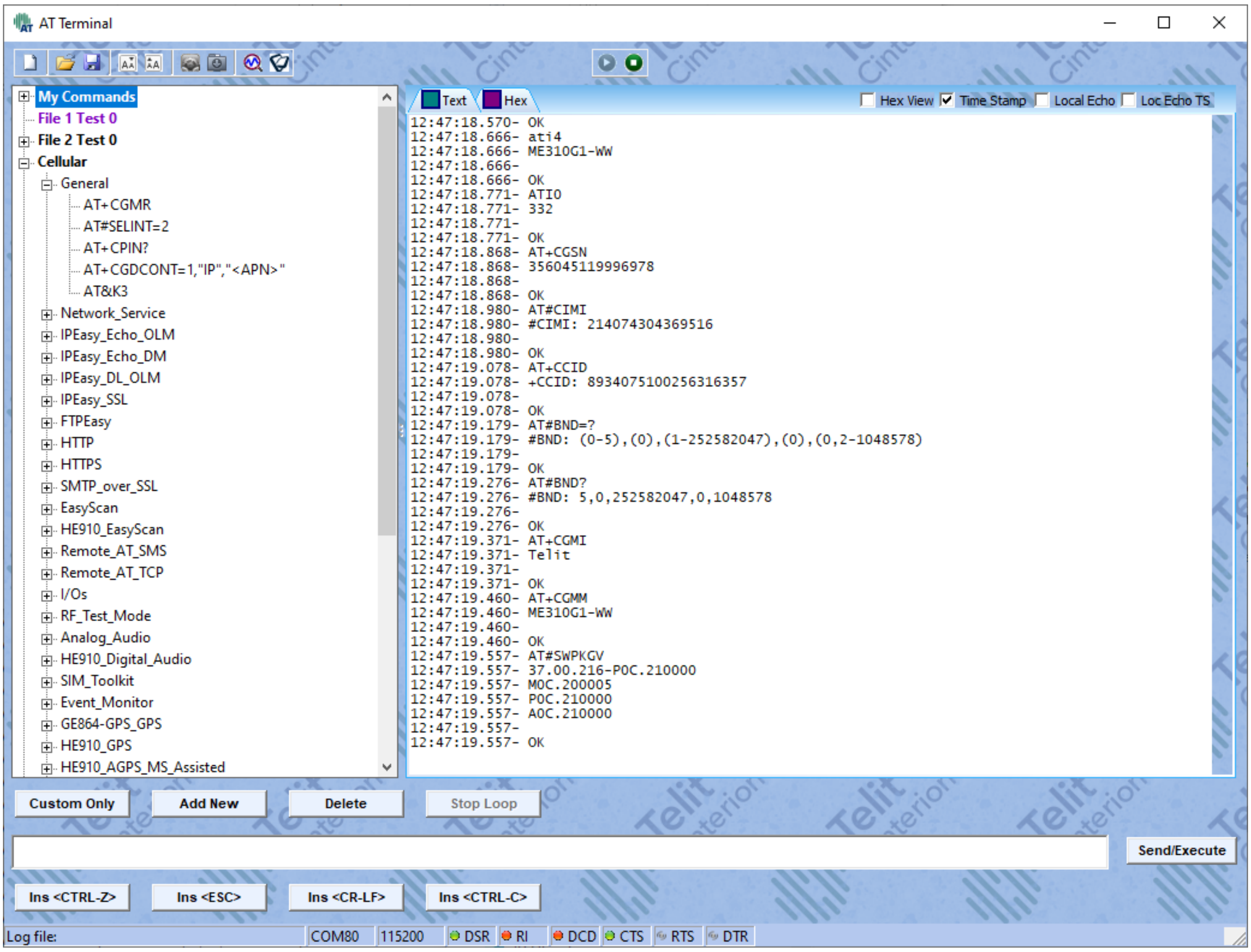

Figure 1 Telit Cinterion AT Terminal

Module registration

Once the module is connected and responsive to AT commands, it is important to verify the registration status with the network. In fact, to establish an MQTT connection, the module must be registered (home or roaming) for data traffic services.

The registration status can be verified by querying the **AT+CREG?**, **AT+CGREG?** (PS services valid for 2G or 3G) and **AT+CEREG?** (EPS services valid for LTE).

Commands example:

- Module registered home network

  AT+CREG?

  +CREG: 0,1

  OK

  AT+CGREG?

  +CGREG: 0,1

  OK

- LTE Module registered roaming

  AT+CREG?

  +CREG: 0,5

  OK

  AT+CEREG?

  +CREG: 0,5

  OK

If the module does not report similar results, please check the “80000NT11696A - 2G 3G 4G Registration Process Application Note”.

PDP context activation

If the module is registered to EPS services (LTE) or PS services (2G/3G), it is possible to obtain an IP address with the AT#SGACT command.

You must select the <cid> related to the APN used for data services.

Commands example:

AT+CGDCONT?

+CGDCONT: 1,"IP","nxt17.net","",0,0,0,0

+CGDCONT: 2,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 3,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 4,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 5,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

+CGDCONT: 6,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

OK

The APN has been configured to cid=1, so the context will be requested for this cid.

AT#SGACT=1,1

#SGACT: 10.35.25.68

OK

3.2 Create a HiveMQ account

In this guide, HiveMQ broker is used to demonstrate how an MQTT connection can be established using a Telit Cinterion module. However, the same procedure can be used to configure the module to establish a connection to a different MQTT broker.

You can create a free HiveMQ account using the link: https://www.hivemq.com/.

Click on "**Get HiveMQ**" to create a free account or to **log in** if you have already an account.

.

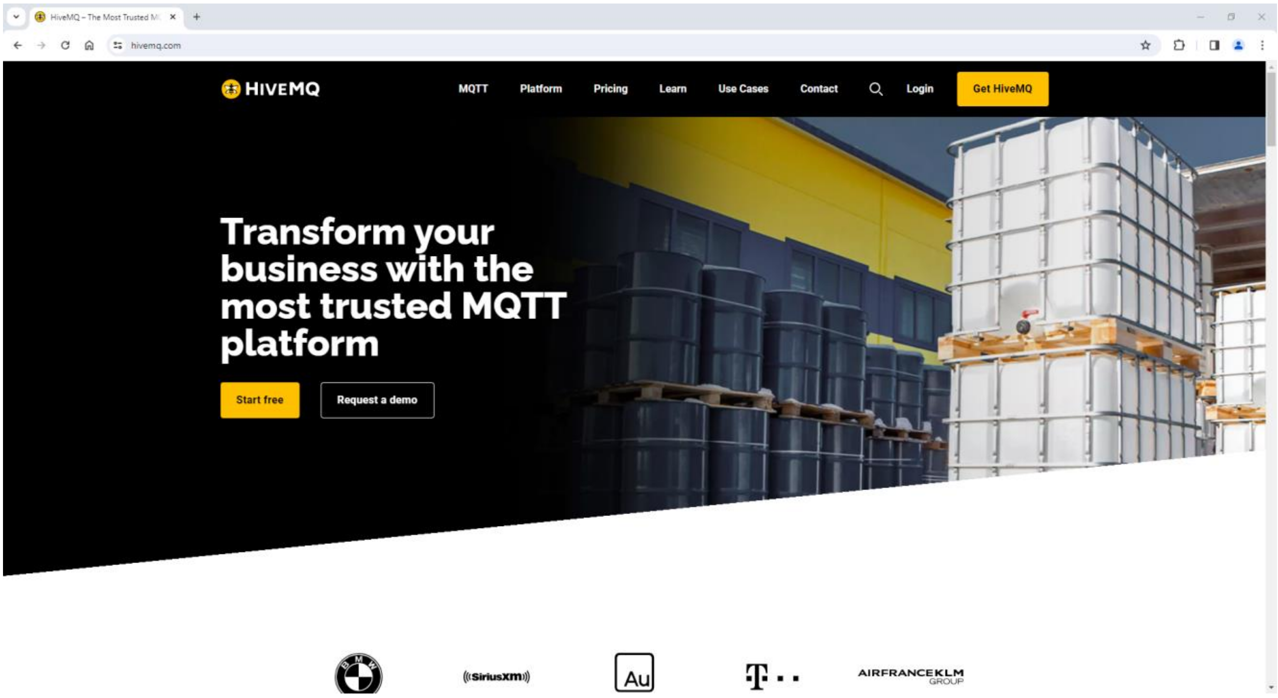

Figure 2 HiveMQ.com Main Page

If you need to create an account, you must select the "**HiveMQ Cloud**" service.

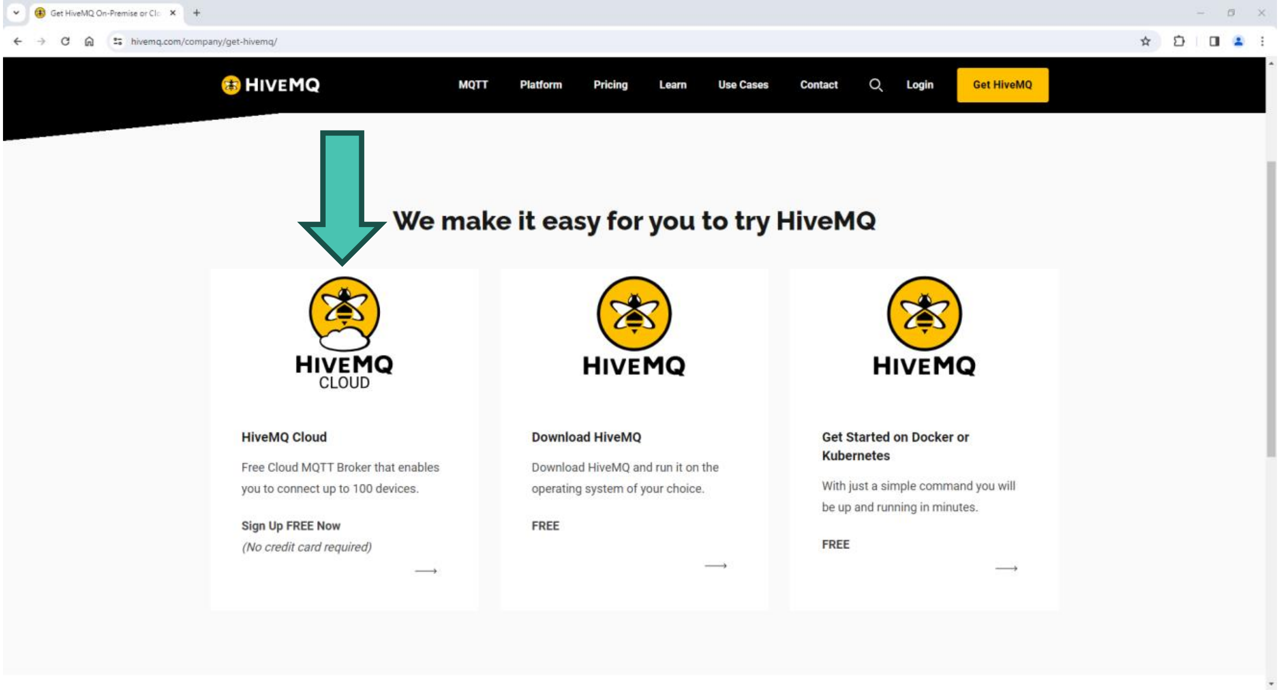

Figure 3 HiveMQ Cloud service

Insert your email and create the password.

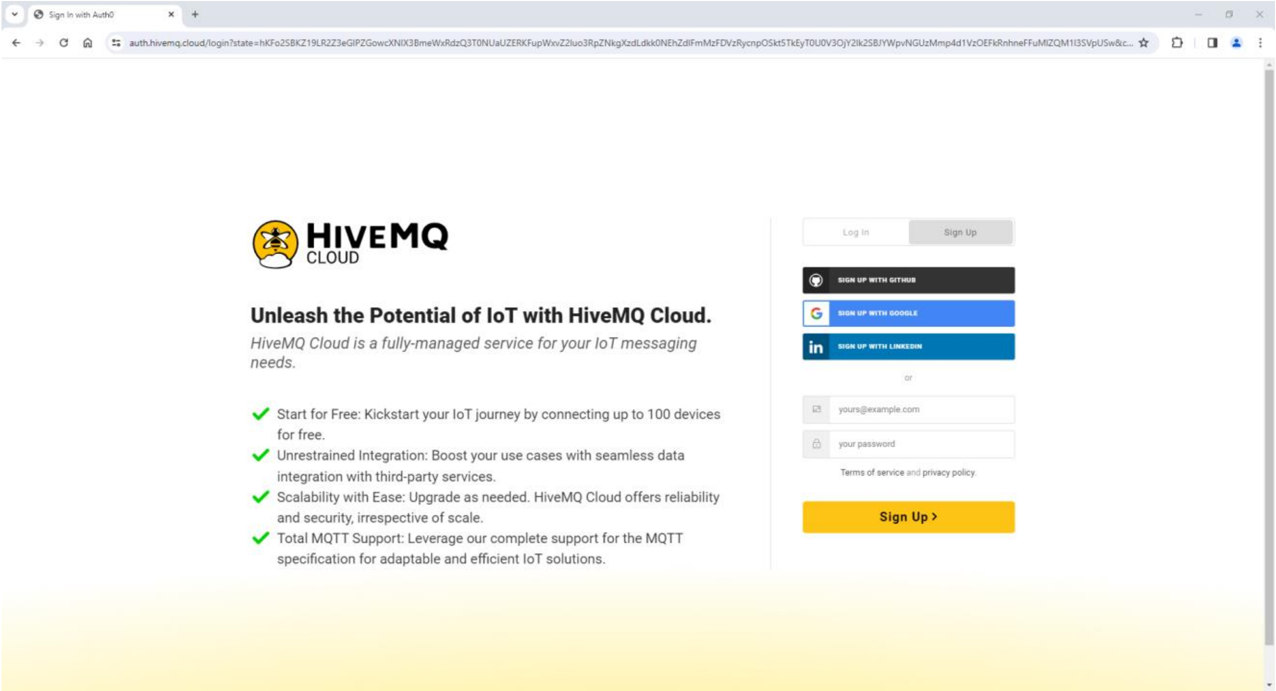

Figure 4 Sign-in page

Complete the profile inserting First name, Last name, Job title, and Company.

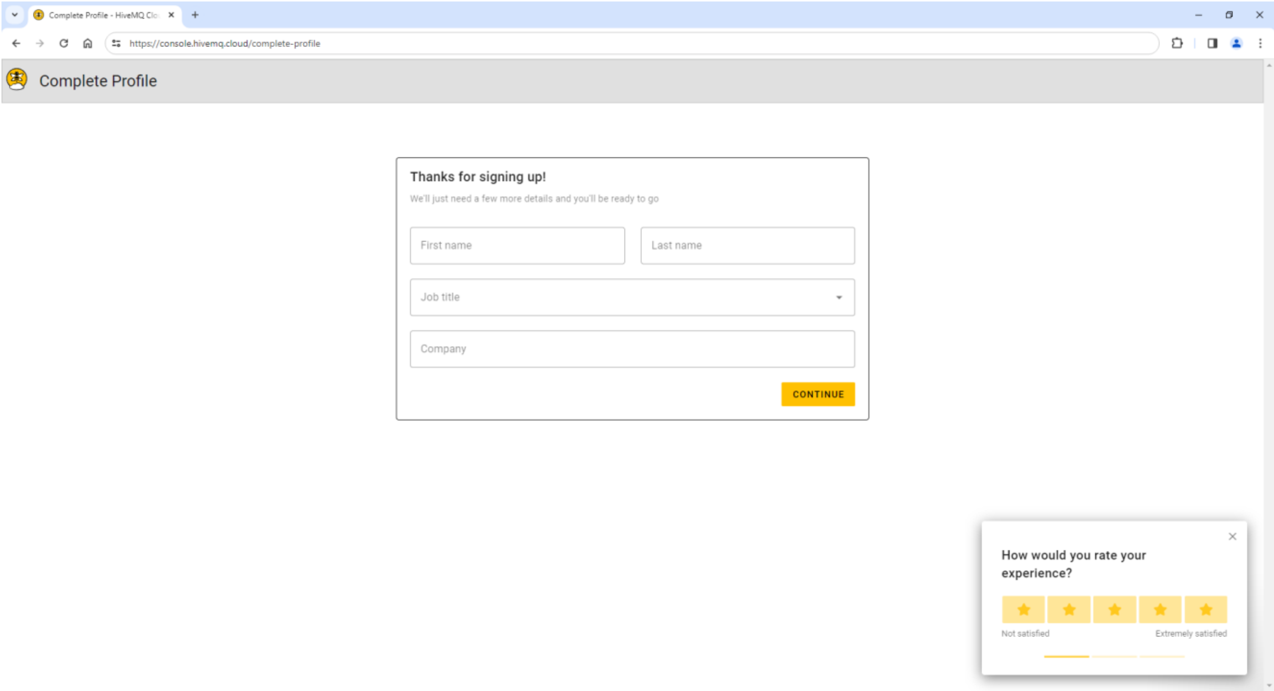

Figure 5 Complete Profile page

Select "**Serverless**" as the HiveMQ Cloud plan. This is totally free and does not require a credit card.

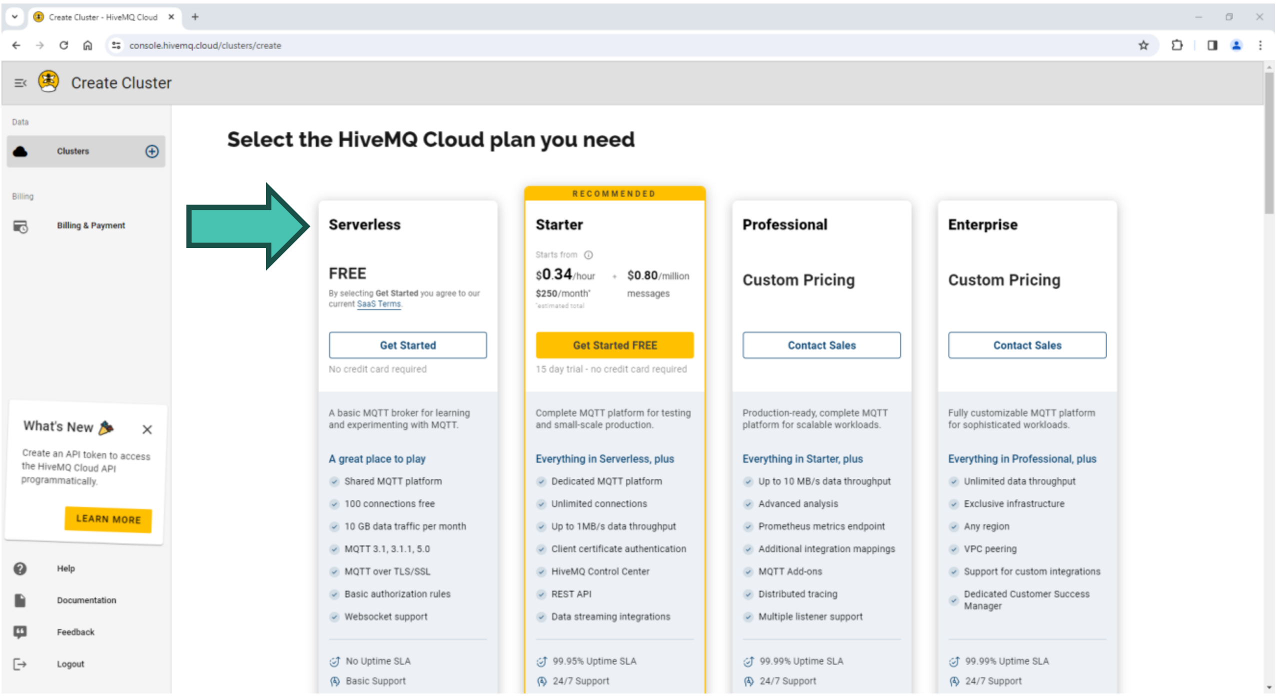

Figure 6 HiveMQ Cloud plan selection

Finally, click on Create to finalize the procedure.

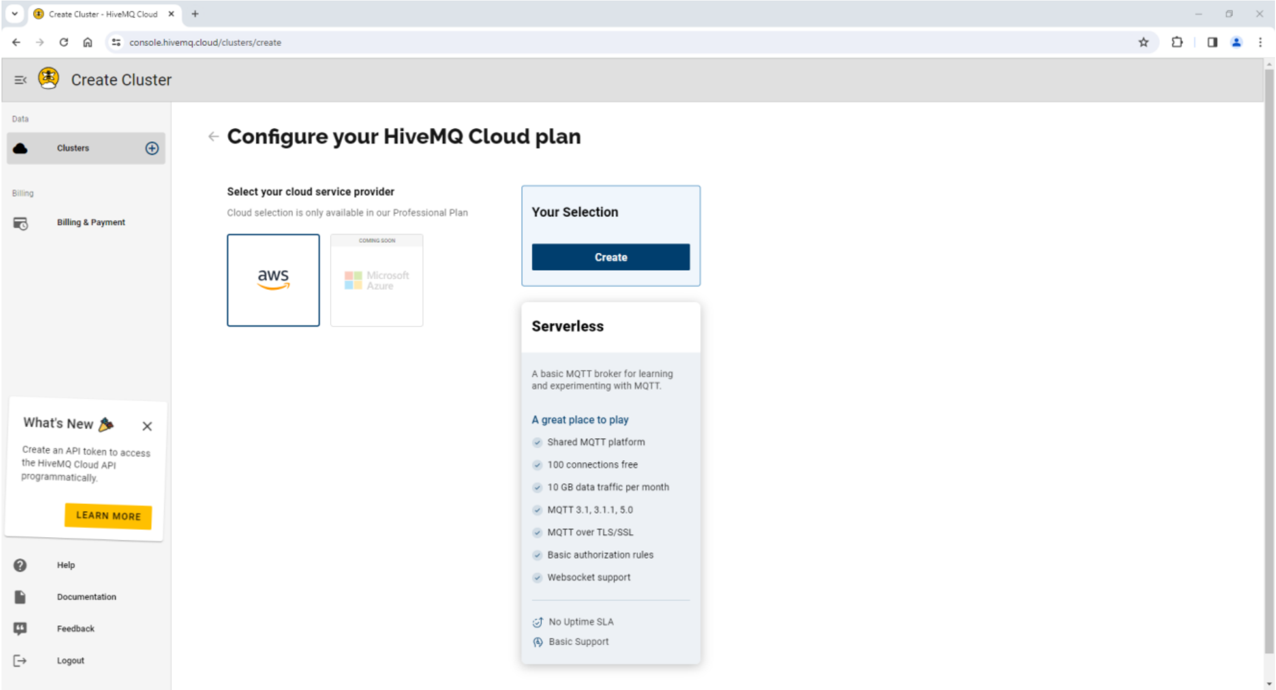

Figure 7 Create Cluster page

3.3 Manage the cluster

All the created clusters are listed in the left panel. Clicking on the “Cluster” button, you will be redirected to a page where all the clusters are represented with single boxes reporting the main information about each of them.

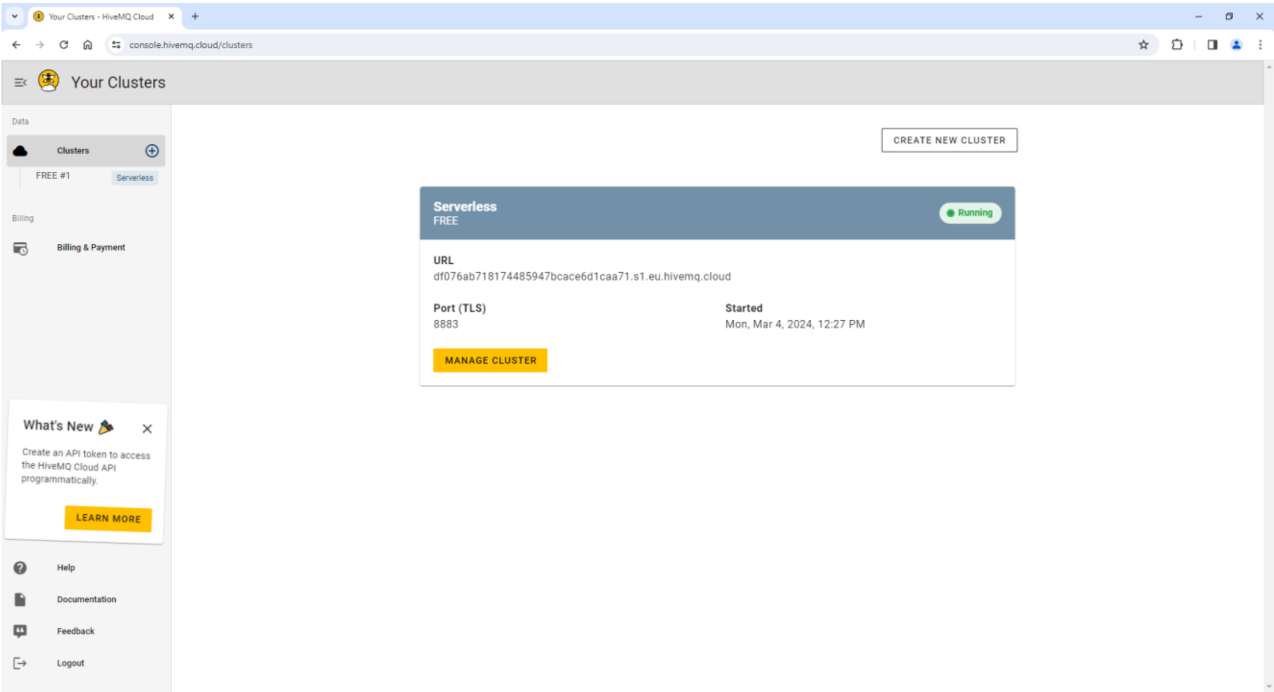

Figure 8 Clusters page

Clicking on “MANAGE CLUSTER” you can modify the cluster’s configuration.

On the next page, you can see a summary of the cluster’s information, the URL and port used to establish the connection, a counter for the MQTT Client sessions and one for the data traffic, and a button to delete it.

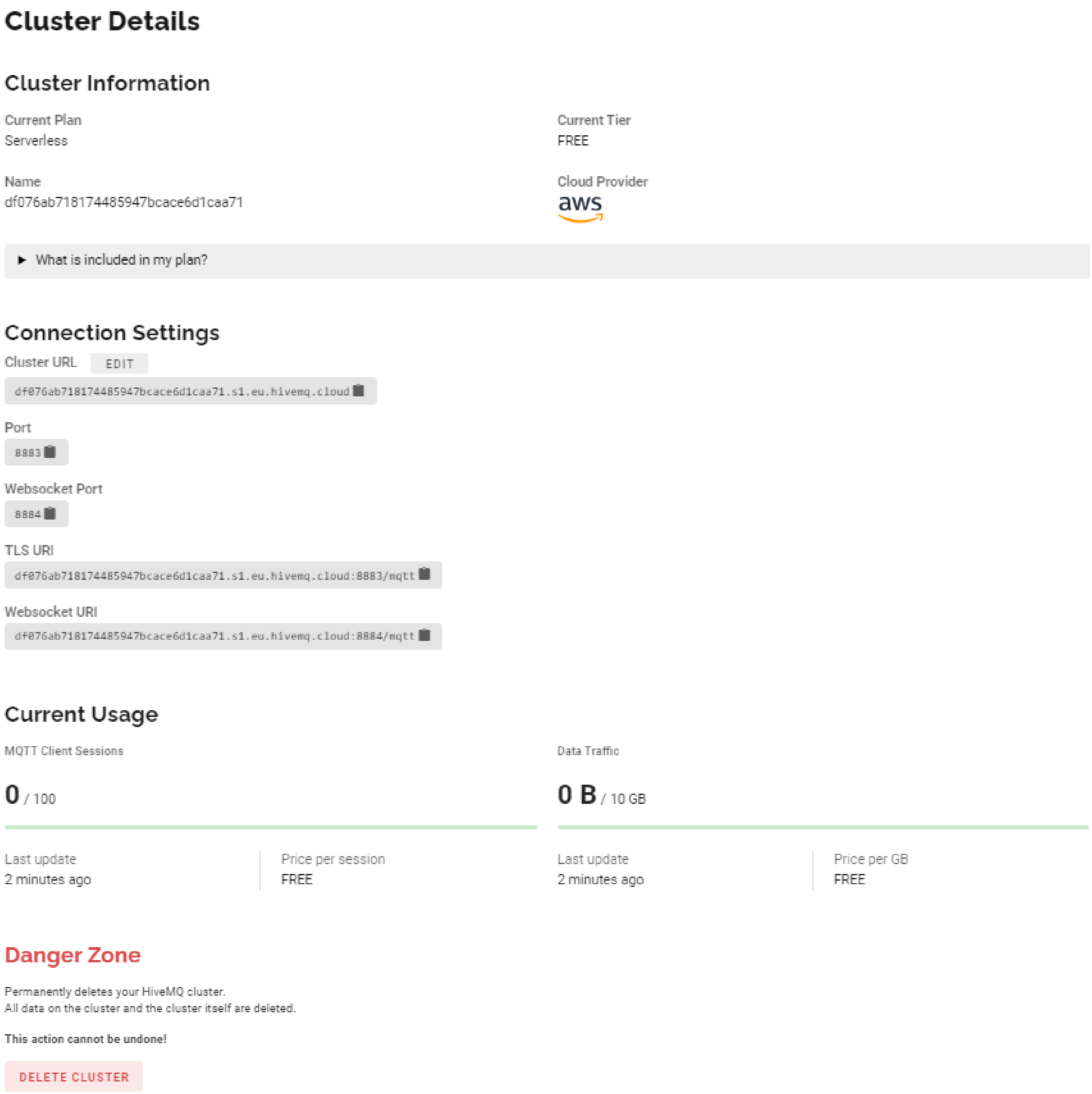

Figure 9 Cluster overview

The "Cluster URL" and the "Port" will be used to establish the connection with the Telit Cinterion module.

3.4 Access Management

To establish an MQTT connection, it is important to add some credentials. These are used by the clients to authenticate themselves to the broker when establishing the MQTT connection.

You can add the credentials, by clicking on the "**ACCESS MANAGEMENT**" tab present in the upper right corner.

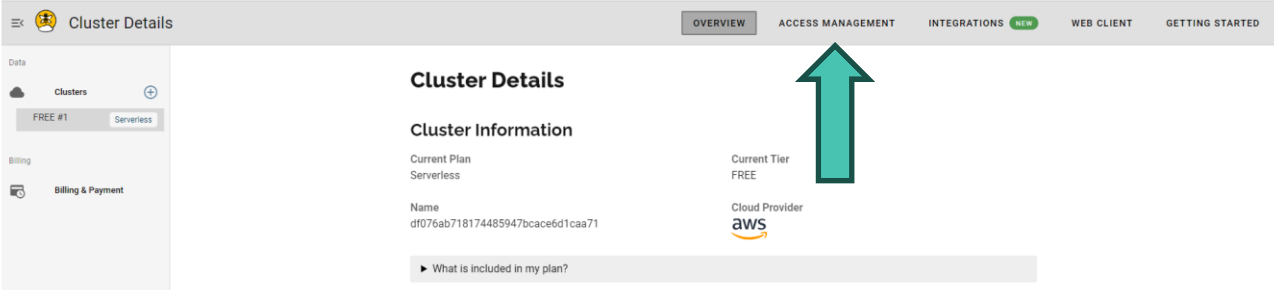

Figure 10 Access Management tab

On the next page, you can add credentials by creating a username, the password, and the permission. The permission tab allows you to select access to subscribe only, publish only, and a combination of these two.

In Figure 11 an example of credential creation is shown, where "**Publish and Subscribe**" permission has been selected.

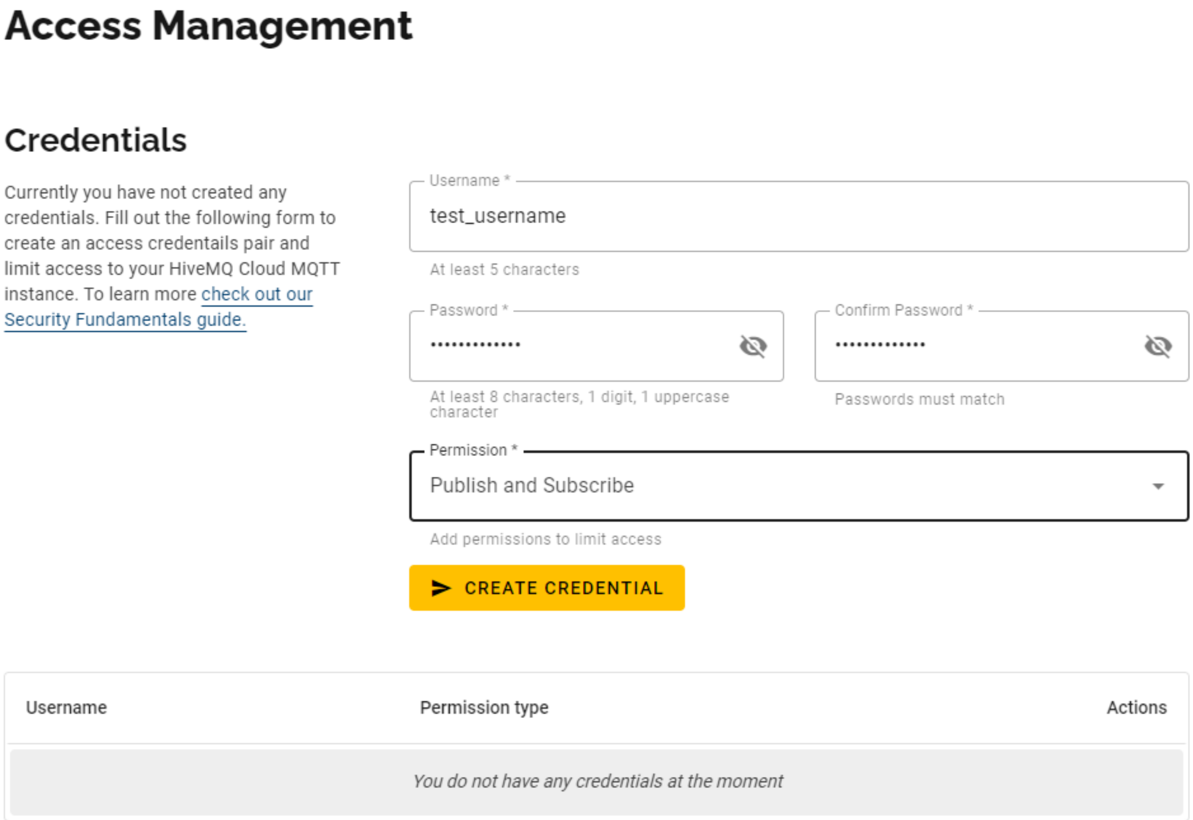

Figure 11 Example of credentials creation

Once inserted the credentials, you must click the "CREATE CREDENTIAL" button to add the user.

A list of all the added credentials is shown in the box below the button.

To delete the credential, you can simply click on the "DELETE" button.

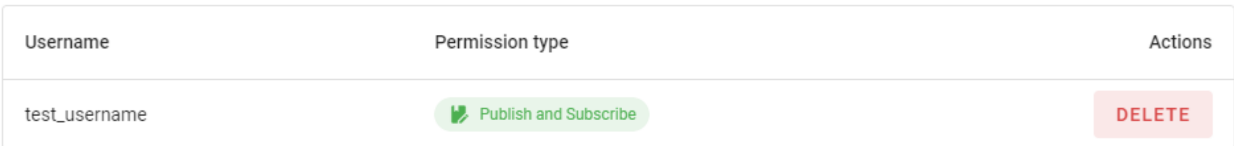

Figure 12 Credentials list

3.5 HiveMQ web client

HiveMQ Cloud provides also a web client, that can be used to monitor graphically the topics present and the messages exchanged.

The web client can be enabled by opening the "**WEB CLIENT**" tab present in the upper right corner. Then you should insert the credentials and click on the "**CONNECT CLIENT**" button.

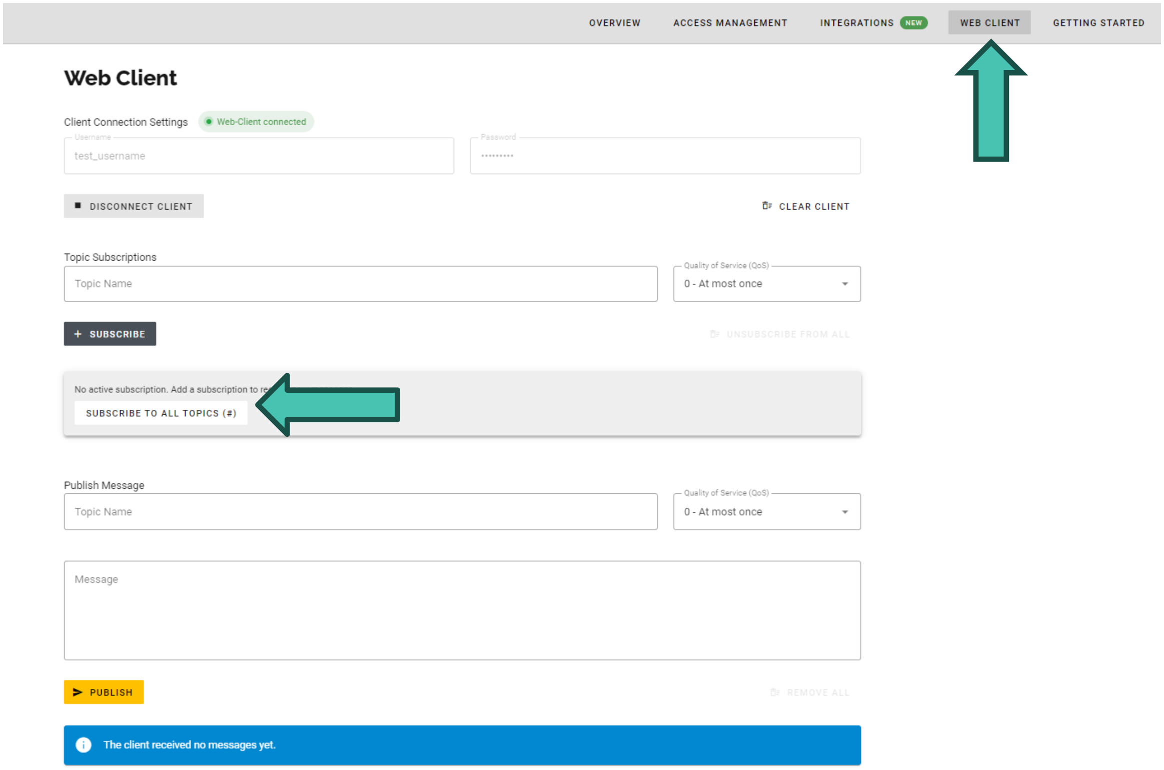

Figure 13 WebClient view

Clicking on the "**SUBSCRIBE TO ALL TOPICS (#)**", the web client can see all the messages sent by all the topics present on the broker. The # wildcard will be explained in 3.10.

3.6 Connect to HiveMQ broker

At this time, the HiveMQ broker has been configured and it is possible to use the Telit Cinterion module to establish the connection. In this tutorial, an ME310G1-WW module will be used.

MQTT Client configuration

The first step is to enable the MQTT client using #MQEN on a specific instance number. In this example, instance number 1 has been chosen.

AT#MQEN=1,1

OK

Then the connection should be configured using the #MQCFG command. In this, <hostname>, <port>, <cid>, and <sslEn> fields can be set.

AT#MQCFG=1,"df076ab718174485947bcace6d1caa71.s1.eu.hivemq.cloud",8883,1,1

OK

Where the URL and port used are respectively “Cluster URL” and “Port” fields reported on the Cluster Details page shown in Figure 9 (“OVERVIEW” tab). Moreover, both <cid> and <sslEn> have been set to 1 since the PDP context used is related to the cid 1 and HiveMQ requires a secure connection.

HiveMQ has a timeout inactivity of 5 minutes. For this reason, can be useful to set a keepalive message with a timeout lower than this period. Let’s set a 4 minutes timeout with the #MQCFG2.

This command allows also to set of a persistent connection. Without a persistent connection, when the client is offline all information and messages that are queued from a previous session are lost.

AT#MQCFG2=1,240,1

OK

The HiveMQ server requires also the usage of Server Name Indication (SNI). This can be enabled by using the #SSLSECCFG2 command.

AT#SSLSECCFG2=1,2,1

OK

Successful connection to the broker

When the MQTT client is completely configured, the connection to the HiveMQ broker can be established using the #MQCONN command.

AT#MQCONN=1,mqtt_client,test_username,Password1

OK

Where the “mqtt_client” is the <clientID> field and identifies each MQTT client that connects to the broker. <userName> and <passWord> fields are the credentials created in subchapter 3.4.

The presence of the client is notified by the counter in the OVERVIEW tab on the HiveMq cloud page.

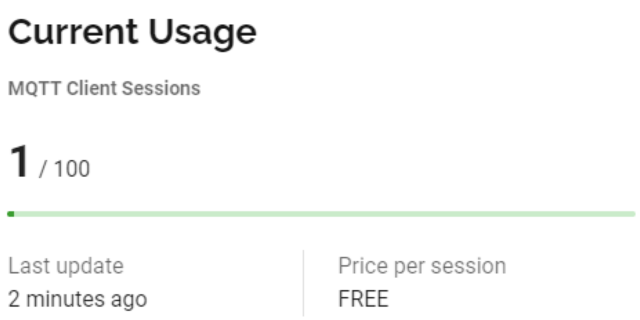

Figure 14 MQTT Client Sessions enabled

3.7 Publish data

Now, we are connected to the HiveMq server and ready to send some data. This is possible by publishing a message on a topic. The topic can be new or already present. Since we have not yet created a topic, we can create a new one with the #MQPUBS command.

AT#MQPUBS=1,testTopic,0,1,new_message

OK

With this command, we have sent the "new_message" to the "testTopic" topic. The message sent is **not retained** by the broker since the <retain> field is equal to zero and it has been sent with QoS equal to 1.

If you are using the web client and you are subscribed to all the topics (as described in 3.5), you can see that the message is received by the broker.

Message	Topic	QoS	Timestamp
new_message	testTopic	0	1709725057253

Figure 15 "new_message" is received on testTopic (Web Client view)

The QoS that is shown in the Web Client is 0 since the Web Client is subscribed to all topics with QoS equal to 0.

3.8 Subscribe to a topic and receive data

When a client is subscribed to a topic, all the messages that are sent to the topic are broadcast to all the subscribed clients.

In 3.7, we have published data on testTopic but we are not subscribed to it. This means that all messages sent to the testTopic (also by other clients) are not received by the module.

It is possible to subscribe the module to testTopic with the #MQSUB command.

AT#MQSUB=1,testTopic

OK

Now, we can try to send a message to the topic.

AT#MQPUBS=1,testTopic,0,1,message2

OK

#MQRING: 1,1,testTopic,8

The #MQRING URC is notifying us that a new message is received by MQTT client 1, which is stored in slot 1, that arrives from "testTopic" and that its length is 8 characters.

Finally, we can read the message using the #MQREAD command, specifying the instance number and the message slot.

AT#MQREAD=1,1

#MQREAD: 1,testTopic,8

<<<

message2

OK

Message sent by the Web Client

The same happens if the message is sent by the Web Client.

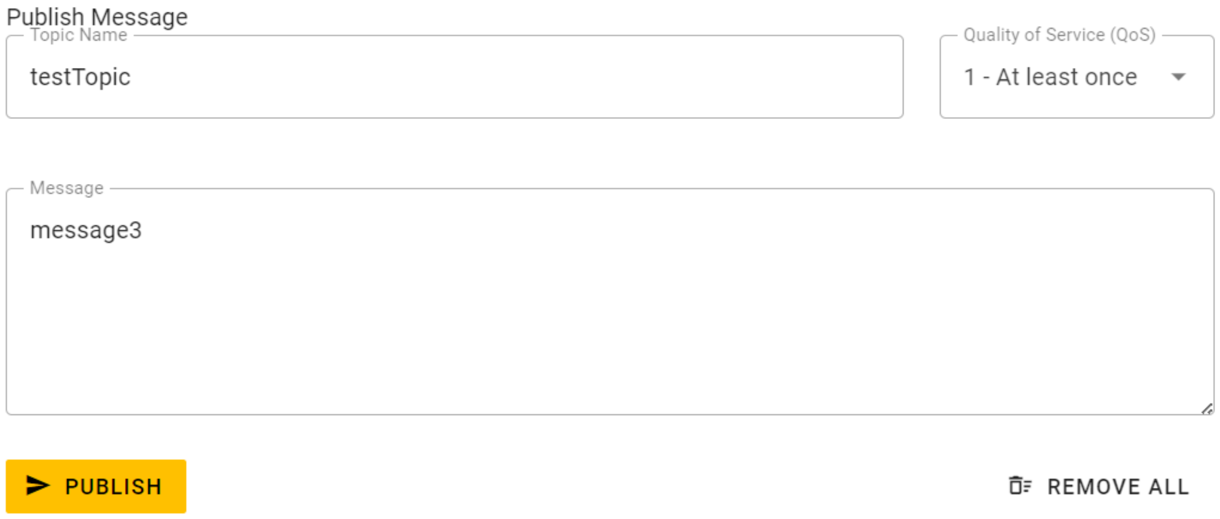

Figure 16 Sending the message from the Web Client

Clicking on the "PUBLISH" button, the message is sent to the topic and received by the module.

#MQRING: 1,1,testTopic,8

AT#MQREAD=1,1

#MQREAD: 1,testTopic,8

<<<

message3

OK

3.9 Retained message

When a message is retained, the broker keeps this message and the corresponding QoS for that topic. Each client that subscribes to that topic will receive the last message retained by the broker.

We can send a retained message to a topic by setting the <retain> field of the #MQPUBS command to 1.

For example, we can send the message "retained_message" to the topic "rtdTopic", and then subscribe to it later.

AT#MQPUBS=1,rtdTopic,1,1,retained_message

OK

AT#MQSUB=1,rtdTopic

OK

#MQRING: 1,1,rtdTopic,16

AT#MQREAD=1,1

#MQREAD: 1,rtdTopic,16

<<<

retained_message

OK

The message is stored by the broker and is received by the module as soon as it subscribes to the topic.

Only the last messages with a retain flag to 1 is kept by the broker.

3.10 Usage of Wildcards

A topic can be single or multilevel. The different levels are separated by a forward slash.

As an example, we can consider the following multilevel topics that refer to different sensors present in different rooms of a house.

myHome/kitchen/temperature

myHome/kitchen/light

myHome/bedroom/temperature

myHome/bedroom/humidity

If we want to subscribe to multiple topics simultaneously, we can use the wildcards.

These can be of two types: single-level and multi-level.

Single level wildcard: +

The single level wildcard is represented by the character “+” and is used to replace all those topics present in a specific level.

For example, if we are interested in receiving only the temperature of all the rooms, we can send the following command.

AT#MQSUB=1,myHome/+/temperature

The client will be subscribed to both myHome/kitchen/temperature and myHome/bedroom/temperature

Multi Level wildcard:

The multi level wildcard is represented by the character “#” and is used to replace all those topics present in a specific level and the lower levels. This can be used only as the last character of the path.

For example, if we are interested in all the sensors present in a room, we can use the following command.

AT#MQSUB=1,myHome/kitchen/#

The client will be subscribed to both **myHome/kitchen/temperature** and **myHome/kitchen/light.**

If instead, we are interested in all the sensors present in the house, we can use it at a higher level.

AT#MQSUB=1,myHome/#

In this case, the client will be subscribed to all the topics: **myHome/kitchen/temperature**, **myHome/kitchen/light**, **myHome/bedroom/temperature,** and **myHome/bedroom/humidity.**

3.11 Publish double quotes nested in a string

In some cases, the message to be sent requires multiple double quotes inside the string (e.g. sending a JSON message). These can be considered as multiple strings by the AT parser and the command parameter can be truncated, reporting in most of cases ERROR. For this reason, it is possible to prevent this problem by using two different approaches.

Using #MQPUBSEXT

The #MQPUBSEXT command allows sending a certain number of bytes specified by the field <dataLen>.

Once you send this command, the “>>” characters are printed and the module enters in online mode expecting the number of bytes to be sent.

AT#MQPUBSEXT=1,testTopic,0,1,36

>>>{"Company":"Telit","Message":"Test"}

OK

#MQRING: 1,1,testTopic,36

AT#MQREAD=1,1

#MQREAD: 1,testTopic,36

<<<

{"Company":"Telit","Message":"Test"}

OK

Using #SETHEXSTR=2

By using AT#SETHEXSTR=2 command, it is possible to insert hexadecimal characters inside a string. With this configuration, we can send strings with nested double quotes when using the #MQPUBS command.

The #SETHEXSTR can be sent at any moment, also after the connection to the broker.

AT#MQPUBS=1,testTopic,0,1,"{\22Company\22:\22Telit\22,\22Message\22:\22Test\22}"

OK

#MQRING: 1,1,testTopic,36

AT#MQREAD=1,1

#MQREAD: 1,testTopic,36

<<<

{"Company":"Telit","Message":"Test"}

OK

To disable the hexadecimal codification the AT#SETHEXSTR=0 must be sent.

3.12 Unsubscribe and close the connection

It is possible to unsubscribe the module with the #MQUNS command.

For example, we can consider the module subscribed to “testTopic”.

AT#MQUNS=1,testTopic

OK

Then we can close the connection with the broker by using the #MQDISC command.

AT#MQDISC=1

OK

Finally, we can disable the MQTT client with the #MQEN command.

AT#MQEN=1,0

OK

4 Acronyms and Abbreviations

Table 2: Acronyms and Abbreviations

Acronym	Definition
ADC	Analog – Digital Converter
CLK	Clock
CMOS	Complementary Metal – Oxide Semiconductor
CS	Chip Select
DAC	Digital – Analog Converter
DTE	Data Terminal Equipment
ESR	Equivalent Series Resistance
GPIO	General Purpose Input Output
HS	High Speed
HSDPA	High-Speed Downlink Packet Access
HSIC	High-Speed Inter Chip
HSUPA	High-Speed Uplink Packet Access
I/O	Input Output
MISO	Master Input – Slave Output
MOSI	Master Output – Slave Input
MRDY	Master Ready
PCB	Printed Circuit Board
RTC	Real-Time Clock
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready
TTSC	Telit Technical Support Centre
UART	Universal Asynchronous Receiver Transmitter
UMTS	Universal Mobile Telecommunication System
USB	Universal Serial Bus
VNA	Vector Network Analyzer
VSWR	Voltage Standing Wave Radio
WCDMA	Wideband Code Division Multiple Access

5 Related Documents

Refer to https://dz.telit.com/ for current documentation and downloads.

Table 3: Telit Related Documents

S.no	Book Code	Document Title
1	80617ST10991A	ME310G1/ME910G1/ML865G1 AT Commands Reference Guide
2	80617NT11957A	MQTT and MQTT-SN Application Note
3	80000NT11696A	2G 3G 4G Registration Process

6 Document History

Table 4: Document History

Revision	Date	Changes
0	2024-03-12	Initial release

From Mod.0809 Rev.9